

The three showings on the claims are: No. 3 on the west boundary of the William claim; the main, lying in the centre; and the southeast, lying in the northeast corner of the William. The positions of the showings are indicated in the plan on page 179, on which sample numbers are also indicated. Sample data and analyses are set forth in the accompanying table.

The No. 3 and the southeast showings are honeycombed quartz veins in quartz diorite. Hand specimens picked from the surface contain no trace of mineralization other than quartz or limonite and some magnetite. No. 3 vein strikes north 50 degrees east and dips 85 to 89 degrees south.

The main vein, on which most of the work has been done, is in a creek near the centre of the group and has been traced nearly 210 feet by rock trenches and stripping. The vein is in a straight definite fracture that strikes north 22 degrees east and dips 66 degrees northwest between walls of andesite slightly mineralized with pyrite. The vein filling is of quartz mineralized with pyrite, chalcopyrite, and secondary bornite.

The vein is mineralized irregularly. Some parts are well mineralized, such as that represented by Sample No. 4. Others, such as that represented by Sample No. 3, are only slightly mineralized.

VANCOUVER ISLAND*

ELK RIVER (50° 127° S.E.)

Iron

Quatsino Copper-Gold Mines Limited

A diamond-drilling and prospecting programme was started in September, 1950. H. L. Hill, consulting engineer for this company, reports that nine diamond-drill holes totalling 1,070 feet were drilled to explore a magnetite deposit and to prospect for copper mineralization along a limestone-greenstone contact.

[Reference: *Minister of Mines, B.C.*, Ann. Rept., 1929, p. 379.]

ZEBALLOS (50° 126° N.W.)

Gold

Privateer Mine Limited.—William Bowen and partners leased the surface dumps and mill of this company. Careful clean-up of the mill and pertinent buildings yielded some concentrates which were shipped to the Royal Mint. Gross content: Gold, approximately 22 oz.

DUNCAN (48° 123° N.W.)

Copper-Zinc

Twin J (Vancouver Island Base Metals Limited)

Company office, Credit Foncier Building, Vancouver. C. Rutherford, manager; C. H. Hewat, mine superintendent. In October, 1950, the Twin J property on Mount Sicker, near Duncan, was reopened. General retimbering and cleaning up was done on the 265- and 300-foot levels. On the 300-foot level 50 feet of drift was driven toward the Richard III section and 60 feet of raise was driven to the 265-foot level. The average number of men employed was twenty.

JORDAN RIVER (48° 124° S.E.)

Copper

Sunloch and Gabbro†

These two adjacent properties are near the settlement of River Jordan on the southwest coast of Vancouver Island, 45 miles by good highway from Victoria. Both were optioned in 1949 from their respective owners, The Consolidated Mining and Smelting

* By R. B. King, except as noted.

† By J. S. Stevenson.

Company of Canada, Limited, and Gabbro Copper Mines Limited, by Hedley Mascot Gold Mines Limited and are being developed as a single operation. The Sunloch property consists of thirty Crown-granted claims and the Gabbro, adjoining the Sunloch on the west, consists of twenty-three Crown-granted claims. All these claims were located between 1915 and 1920. In 1949 Hedley Mascot located ten claims and a fraction adjoining the Gabbro claims on the west.

Access

The workings are reached by a branch road, 1 mile long, that leaves the Victoria highway about half a mile east of River Jordan post office and extends to within a mile of the workings. From that point the grade for a railway track (24-inch gauge) leads past the Gabbro showings to the Sunloch showings and adits. The railway is now in disrepair, but the grade makes a good trail and with a moderate amount of work could be restored.

Development work on these properties includes numerous open-cuts and strippings, about 3,800 feet of adits, and 14,000 feet of diamond-drilling. The River adit, elevation 609 feet, and the Cave adit, elevation 608 feet, on the east side of the river, and the Centre, elevation 549 feet across the river from them, are on the Sunloch Nos. 5 and 6 claims. There are on the Gabbro two short adits—one, the Winkler, 83 feet long, at elevation 570 feet, on the Vulcan No. 2 claim adjacent to the old railway grade, and the other, the Hornet, 12 feet long, on the east bank of Sinn Fein Creek, about 100 feet northwesterly from the northwest corner of the Black Hornet claim.

The B.C. Electric Railway Company forebay at the end of the flume-line, at an elevation of 1,100 feet, about 550 feet above the principal Sunloch workings, covers about 9 acres and is used to control the flow of water through the penstocks into the power-house at sea-level at the mouth of Jordan River. This power-house generates electricity for the city of Victoria. A branch power-line from the power-house leads up the hillside and, at its closest point, is half a mile east of the Sunloch adits.

History

The discovery of copper on southern Vancouver Island dates back to 1863, when it was found at East Sooke by Capt. Jeremiah Nagle. Little work was done on the discoveries until World War I, when, between 1915 and 1918, 1,940 tons of ore, yielding 177,613 pounds of copper, is recorded as having been shipped from the two principal properties, the Willow Grouse and Copper King at East Sooke. In 1915, at the beginning of this general period of copper activity at Sooke, George Winkler, now of Victoria, made the first discovery of copper on the Sunloch at Jordan River. He was led to the discovery by finding chalcopyrite float on the beach near the mouth of the Jordan River. He succeeded in tracing the float up the difficult canyon to its source in inconspicuous outcrops of the Cave and River zones, on the steep east wall of the canyon. Further prospecting by Mr. Winkler in subsequent years resulted in finding the several other mineralized zones both on the Sunloch and Gabbro properties.

In 1917 Winkler bonded the property to the Sunloch Mining Company, which built the narrow-gauge railway, drilled the first diamond-drill holes, and did the first underground work.

In 1919 The Consolidated Mining and Smelting Company acquired control of the Sunloch Company and continued diamond drilling and underground work. Operations were suspended in 1920. By that time the River, Centre, and Cave adits totalling 3,776 feet had been driven, and holes amounting to 3,470 feet had been drilled on the Sunloch property.* On the Gabbro property the Winkler adit and the Hornet adit had been driven by 1920, and surface exploration was done in the next few years.

* *Minister of Mines, B.C., Ann. Rept., 1920, pp. 220, 221.*

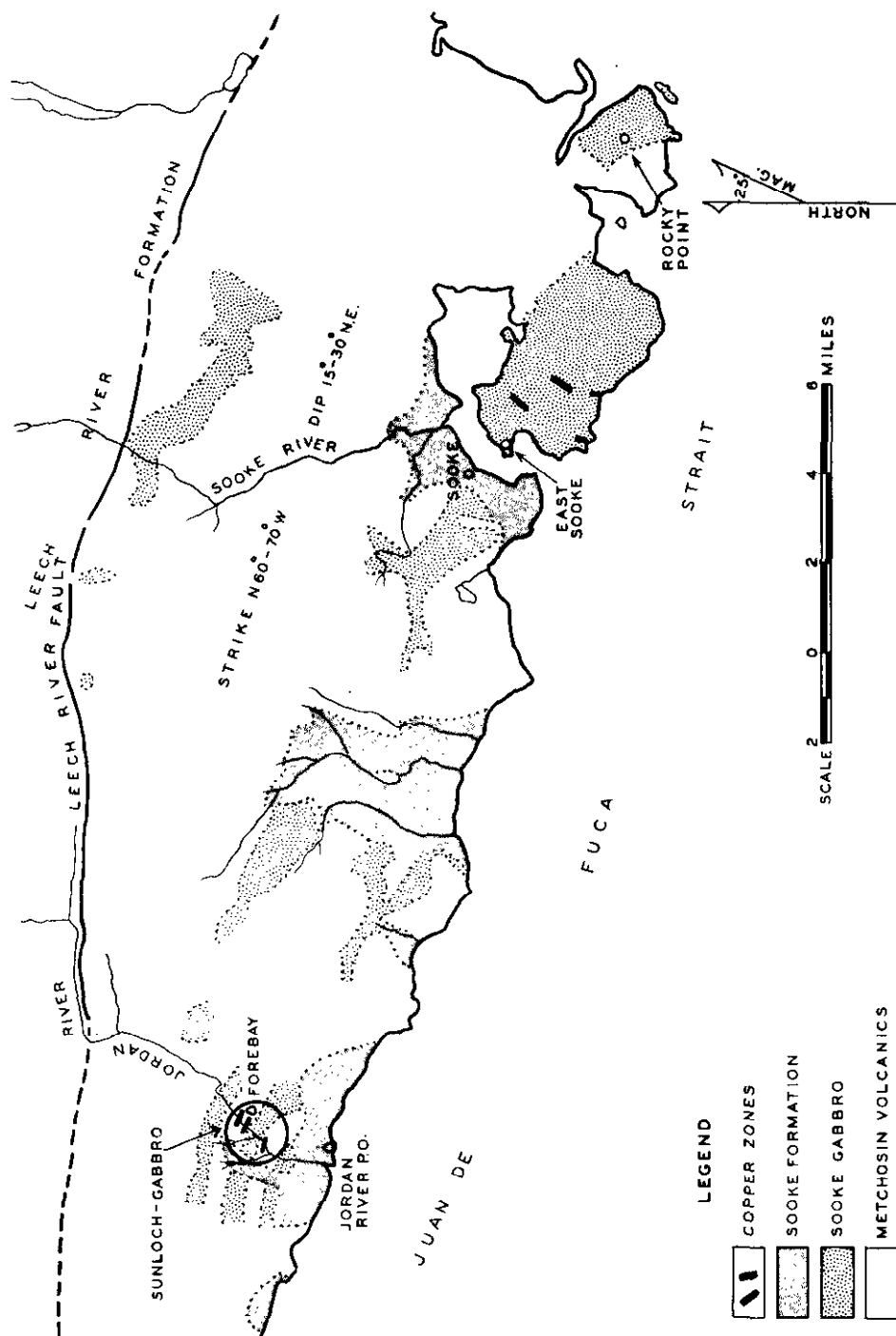


Fig. 9. Geology of Sooke-Jordan River area. Geology adapted from Sooke and Duncan sheets, Geological Survey, Canada, and company plans.

No further work was done on these properties until 1949, when Hedley Mascot Gold Mines Limited optioned the Sunloch property from The Consolidated Mining and Smelting Company and the Gabbro property from Gabbro Copper Mines Limited. Work done by Hedley Mascot on the two properties has consisted principally of diamond drilling, of which 9,354 feet was reported for 1949 and 4,082 feet for 1950. Twenty-eight holes, Nos. 15 to 36 and X-1 to X-6, aggregating 9,354 feet, were drilled before exceptionally high water in Jordan River in November, 1949, flooded desirable drilling-sites and forced suspension of drilling. Drilling was resumed in the spring of 1950, and Holes No. 37, length 1,196 feet, and No. 38, length 1,257 feet, had been drilled when the property was last examined. Subsequently, Hole No. 39 was completed; Hole No. 40 was begun and has been completed since the end of 1950. No underground work has been done by Hedley Mascot on either property.

Production

No production has yet been made from the Sunloch and Gabbro properties. However, The Consolidated Mining and Smelting Company stored several hundred tons of development ore behind a cribbing at the portal of the Cave adit on the Sunloch.

General Statement

The writer, with one assistant, spent two weeks in the autumn of 1949 and two weeks in the summer of 1950 on the Sunloch and Gabbro properties and, in the preparation of this report, studied 115 thin sections and twenty-four polished sections under the microscope.

The kindness of the officials of Hedley Mascot Gold Mines Limited in providing full access to their maps, plans, and other information is gratefully acknowledged. In particular, the writer would like to thank J. W. Young, company geologist in charge of development work on the properties, for his many helpful suggestions concerning the geology of the properties. The writer also wishes to thank George Winkler for directing his attention to many interesting features on the Sunloch and Gabbro properties and for much information concerning their history and development.

Topography

The Sunloch and Gabbro properties are in an area of rugged topography. The Jordan River flows through the important, central part of the properties in a rocky canyon about 500 feet deep and a mile and a half long. The showings outcrop in this canyon, and the adits have been driven from one or the other of its walls at heights ranging from 15 feet to 90 feet above the normal level of the river. The river level is apt to rise 5 feet or thereabouts in a few hours during heavy rains, particularly in the autumn. Upstream from the canyon, on either side, the ground, although not precipitous, is steep.

At one time the area was heavily forested, but much of the timber has been cut from it. However, some Douglas fir and considerable hemlock still remain in and adjacent to the canyon. Outcrops are numerous and extensive in the bed of the river and on the canyon walls, but away from the canyon the surface is completely covered by a mantle of black muck and drift, and bedrock does not outcrop except in parts of creek beds. Because of this, little prospecting has been done away from the canyon.

General Geology

The general geological setting of the area of the Sunloch and Gabbro properties may best be considered in its relation to that of the larger Sooke-Jordan River area (Fig 9), of which it is a part.

The oldest rocks in the area include the Leech River formation, a series of argillites and sandstones that have been metamorphosed into slaty and quartzose schists. These

rocks strike easterly and have been closely folded so that most of the strata have very steep dips. They are in contact on the south, along the Leech River fault, with the next youngest formation, the Tertiary Metchosin volcanics.

The Metchosin volcanics underlie a belt 5 to 10 miles wide which extends west-northwest across the southern end of the Island. The volcanics consist principally of basalt with some diabase. Near the Sunloch and Gabbro workings they include porphyritic and non-porphyritic, amygdaloidal varieties; beyond these properties, well-developed pillow lavas, flow breccias, and fragmental types are found. They strike from north 60 to 70 degrees west and dip 15 to 30 degrees northeastward, although in several places they may be vertical or dip steeply southward. Based on fossils found in interbedded tuffs, these volcanics are considered to be of upper Eocene age (Clapp and Cooke, 1917, p. 290); they are not to be correlated with other Tertiary volcanics on the mainland of the Province, which are Oligocene and Miocene. The Sooke-Jordan River area is the type locality for the Metchosin volcanics. Extensive areas of these volcanics have been described from the Olympic Peninsula southerly across Juan de Fuca Strait.

Tertiary marine sandstones and conglomerates fringe the western and southern coasts of Vancouver Island, and in this area they have been named the Sooke formation. These sediments tend to lie in coastal basins, most of which are drained by small streams flowing into the ocean. The Sooke sediments strike westerly (parallel with the coast) and dip from 2 to 5 degrees southerly (offshore). The age of the Sooke formation has been determined as middle Tertiary, probably belonging to the upper portion of the lower Miocene (Clapp and Cooke, 1917, p. 339).

The most unique feature of the geology of the Jordan River-Sooke area is the Tertiary Sooke gabbro, to which the copper mineralization of Jordan River and East Sooke appears to be genetically related. This gabbro occurs as several stock-like masses that trend northwesterly across the southern tip of the Island. The largest of these masses is at East Sooke. This mass is elliptical, with a major axis about 5 miles long and a minor axis about 2½ miles; other masses range down to those about a mile in diameter. Although most of the gabbro areas are circular to elliptical and are probably stocks, those on the Sunloch and Gabbro properties are definitely elongated, striking with the enclosing rocks, and may be sills. The gabbro intrudes the Metchosin basalts of upper Eocene age and is overlain by the Sooke sedimentary formation of early Miocene age, or possibly middle or upper Oligocene. Its age is therefore probably lower Oligocene (Clapp and Cooke, 1917, p. 304).

A second feature in the area of some significance is the Leech River fault. This is a reverse or thrust fault, strike easterly and dip 45 to 75 degrees northward, that extends across the full width of the Jordan River-Sooke area. Clapp (1917, p. 279) summarizes his account of this fault as follows:—

“The Leech River fault is a reverse or overthrust, nearly a strike and nearly a dip slip fault, it is of great length, at least 40 miles and possibly 140 miles long; throughout its known and assumed length it separates the comparatively unmetamorphosed Tertiary rocks from the pre-Tertiary metamorphic and plutonic rocks which lie to the north and have been pushed up over the Tertiary rocks.”

The Leech River fault is accompanied by zones of slaty schists and sheared and schistose volcanics as much as 300 feet or more in width. Southerly from the fault, within the area underlain by Metchosin volcanics, other westerly trending zones of shearing and schistose rocks have been found. These include northwesterly trending zones, 300 and 200 feet wide, of schistose rock that may be seen along the B.C. Electric flume-line 4,000 and 5,000 feet respectively from the forebay. Because of the similarity of the rocks on either side of these shear zones, the nature of the displacement cannot be discovered. These shear zones, and those in which the copper deposits are found, strike northwesterly, and may be genetically related to the Leech River fault.

Detailed Geology

The copper deposits on the Sunloch and Gabbro properties occur in shear zones in the Metchosin volcanics close to a sill-like mass of Sooke gabbro (Fig. 10). Diabase dykes cut the gabbro and may also cut the Metchosin volcanics, but do not appear to be related to the copper mineralization.

Metchosin Volcanics.—Near the copper deposits, the Metchosin volcanics are predominantly basalt. As the canyon provides an abundance of well-washed outcrops and much diamond drilling has been done in the volcanics, a good opportunity is afforded to study the basalt and particularly the textural variations that occur. The basalt is fine-grained, dark greenish grey, and under the microscope is seen to consist of plagioclase and dark green hornblende; no augite was seen.

Although all the basalt in the canyon workings and drill core is partly altered to a hornblende rock, and much of it completely altered, several textural types may be recognized. An amygdaloidal type consists of basalt in which white amygdules occur either sparingly or very close together. The amygdules are so close in places that the rock must have been pumiceous when deposited. The zones in which the amygdules are closely packed together are rarely more than a few inches thick, whereas the zones of the more

SAMPLING DATA ON BALSAM AND HEMLOCK ON THE SUNLOCH AND GABBRO PROPERTIES

Sample No. (Parts A and B)	Notes	P.P.M. in Green Twigs				Copper:Zinc Ratio	
		Copper		Zinc			
		A ¹	B ²	A	B	A	B
1	River zone.....	2.4	1.2	11.2	8.4	0.21	0.14
2		4.0	3.2	10.6	10.6	0.38	0.30
3		3.2	—	15.3	—	0.21	—
4	Hornblendized basalt.....	1.4	1.6	13.4	11.7	0.10	0.14
5		3.2	3.4	11.5	16.7	0.28	0.20
6	Centre zone.....	6.4	6.8	11.2	17.9	0.57	0.38
7		2.4	—	15.6	—	0.15	—
8	Hornblendized basalt.....	2.0	3.8	16.2	12.3	0.12	0.31
9		4.8	5.6	12.3	10.0	0.39	0.56
10	Cave zone.....	6.2	4.6	11.7	12.8	0.53	0.36
11		5.6	8.4	2.7	14.2	2.10	0.59
12		15.6	7.6	11.2	11.2	1.40	0.68
13		11.7	—	18.5	—	0.63	—
14		—	5.15	—	14.6	—	0.35
15	Hornblendized basalt.....	5.0	—	15.9	—	0.31	—
16		3.8	21.4	14.5	10.0	0.26	2.10
17		4.0	—	17.0	—	0.24	—
18		8.2	7.5	23.7	—	0.36	—
19		4.9	—	27.0	—	0.18	—
20		4.4	4.8	25.0	21.3	0.19	0.22
21	Gabbro.....	3.1	4.9	22.0	16.0	0.14	0.30
22		4.4	3.6	30.8	11.8	0.14	0.31
23		4.9	6.2	16.5	24.8	0.29	0.25
24	Winkler zone.....	8.8	10.6	15.3	18.9	0.57	0.55
25	Tiger zone.....	35.2	44.0	11.8	15.4	2.98	2.88
26		11.0	15.0	15.3	13.0	0.72	1.15
27	Gabbro talus.....	6.6	7.05	30.3	28.4	0.22	0.25
28	Larger trees, 8-inch base, at gravel pit, ½ mile southwest from Sample No. 27.....	6.15	9.7	45.0	32.0	0.14	0.30

¹ Balsam.² Hemlock.

widely spaced amygdulæ may be as much as 3 feet thick. The amygdulæ are commonly of plagioclase, but hornblende, chlorite, and clinozoisite may also occur. A porphyritic type of basalt is moderately widespread. In this type the phenocrysts are widely scattered plagioclase crystals about one-sixteenth of an inch long; however, zones several feet wide, in which the phenocrysts are small hornblende crystals, are occasionally seen. Perhaps the most widespread variety of basalt is a diabasic type that consists of distinguishable feldspar laths, one thirty-second of an inch long, intergrown closely enough to give a diabasic texture to the rock. Although the basalt has been intruded by the gabbro, the texture at or near the contacts appears to have been little affected by heat from the gabbro.

Most of the basalt on the property is massive and possesses few features that may be used in determining the structure of the lava. Pillow lavas and flow breccias useful in such studies are absent, and only a few flow contacts were seen. A flow contact between a medium-grained diabasic phase and a fine-grained phase is indicated at a depth of 266 feet in the core from Drill-hole No. 21, and in Drill-hole No. 23 a flow contact between a fine-grained diabasic phase and a phase characterized by indistinct hornblende phenocrysts is indicated at a depth of 291 feet. At a point on the east bank of the river, about 500 feet downstream from the River adit, a contact between porphyritic basalt and fine even-grained basalt was seen to strike north 10 degrees east and to dip 70 degrees north-westward. This indicates local northerly deviations of the strike and reversals in dip from the general northwesterly trend and northeasterly dips seen elsewhere in the area.

Gabbro.—Three northwesterly trending bands of gabbro occur on the property. These bands range in width from 500 to 3,000 feet, are separated by about 3,000 feet of basalt, and are known to extend along their strike for about 4 miles. The centre band, from 2,000 to 3,000 feet wide, is the widest of the three. As the copper ore on the property occurs in mineralized shear zones in the basalt along both contacts of this band of gabbro, it appears to be the most important of the three economically, and has therefore been more carefully studied than the other two bands. Further discussions of gabbro in this report will refer only to this centre band. The rock cuts along the old railway grade and the numerous outcrops in the canyon, both of which cut across the centre band, afford a good opportunity to study the gabbro from one contact to the other. As no diamond drilling has been done in the gabbro, no core from this rock is available for study.

The gabbro is a dark greenish-grey coarse-grained rock with conspicuous plagioclase crystals one-eighth to one-quarter of an inch long. The ferromagnesian mineral in the normal gabbro elsewhere in the Jordan River-Sooke area is principally augite, but on the Sunloch and Gabbro properties it is principally hornblende. However, some relicts of the primary augite remain in this gabbro as cores within crystals of secondary hornblende.

The gabbro is massive and, lacking linear or parallel alignment of either inclusions or constituent minerals, shows no evidence of flowage. In detail the mass is not homogeneous; it includes both medium- and coarse-grained phases and in places includes patches, 6 inches to several feet across, of white altered gabbro. These patches stand out in marked contrast to the predominant dark-green unaltered gabbro. Under the microscope, rock from the patches of white gabbro is seen to consist largely of scapolite and some hornblende. The scapolite has formed by hydrothermal alteration of plagioclase. Alteration of the augite to hornblende and the local alteration in the white patches of plagioclase to scapolite appear to be the only effect that hydrothermal solutions have had on the gabbro.

The northeastern contact of the gabbro with the basalt is not exposed either in the canyon or on the railway grade, but the rock within 50 feet of the probable contact is normal, relatively unaltered gabbro. The southwestern contact may be seen in the canyon. This contact is not sharply defined, but is a zone, about 50 feet wide, that consists of mixed gabbro and basalt. The basalt in the contact zone possesses a definite hornfels texture, characteristic of recrystallization by heat, a consequence of the intrusion of the gabbro into the basalt.

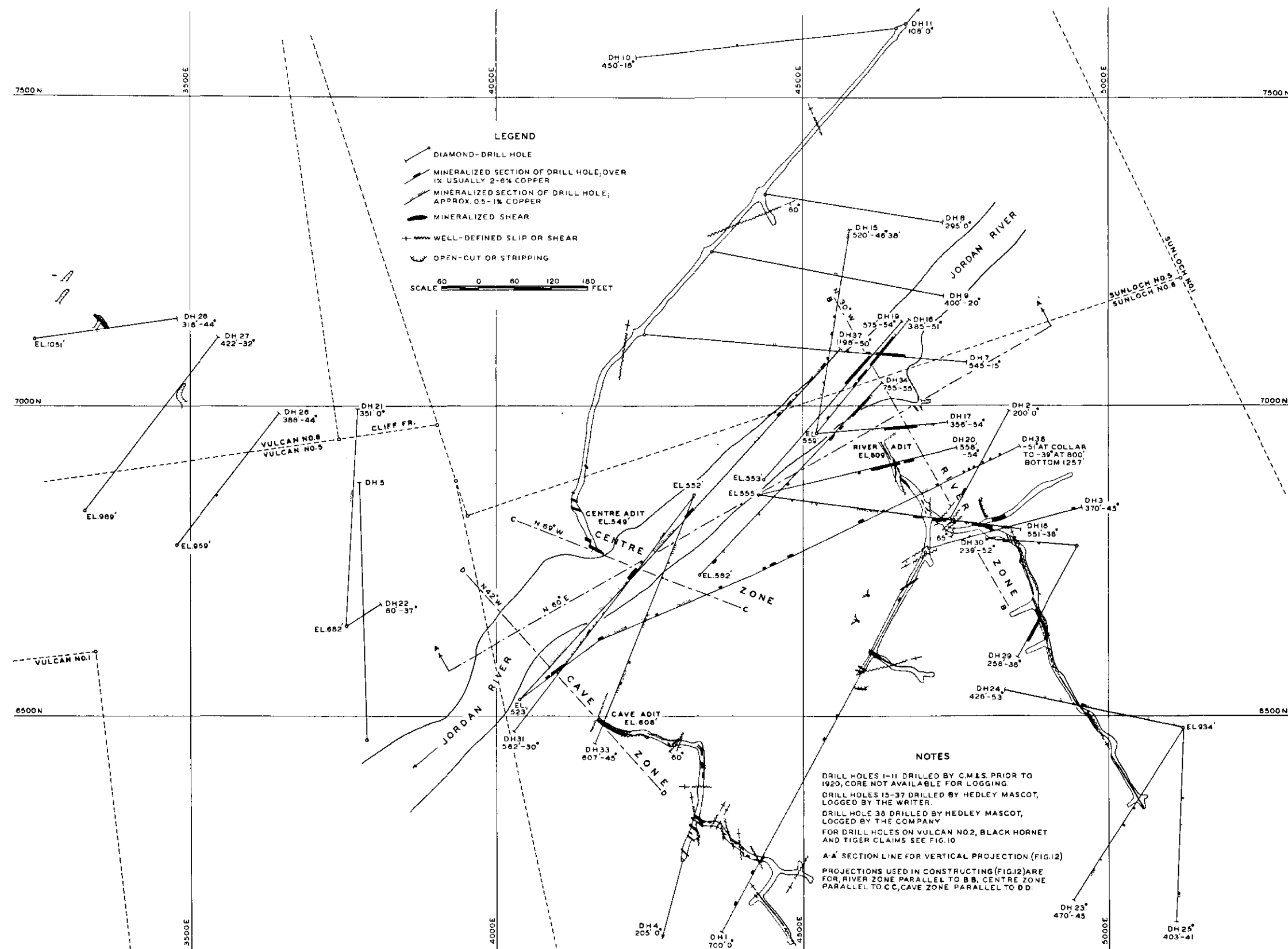


FIG.11. SUNLOCH-GABBRO. DETAILS OF MINERALIZATION IN ADITS AND DIAMOND-DRILL HOLES ON RIVER, CENTRE AND CAVE ZONES.

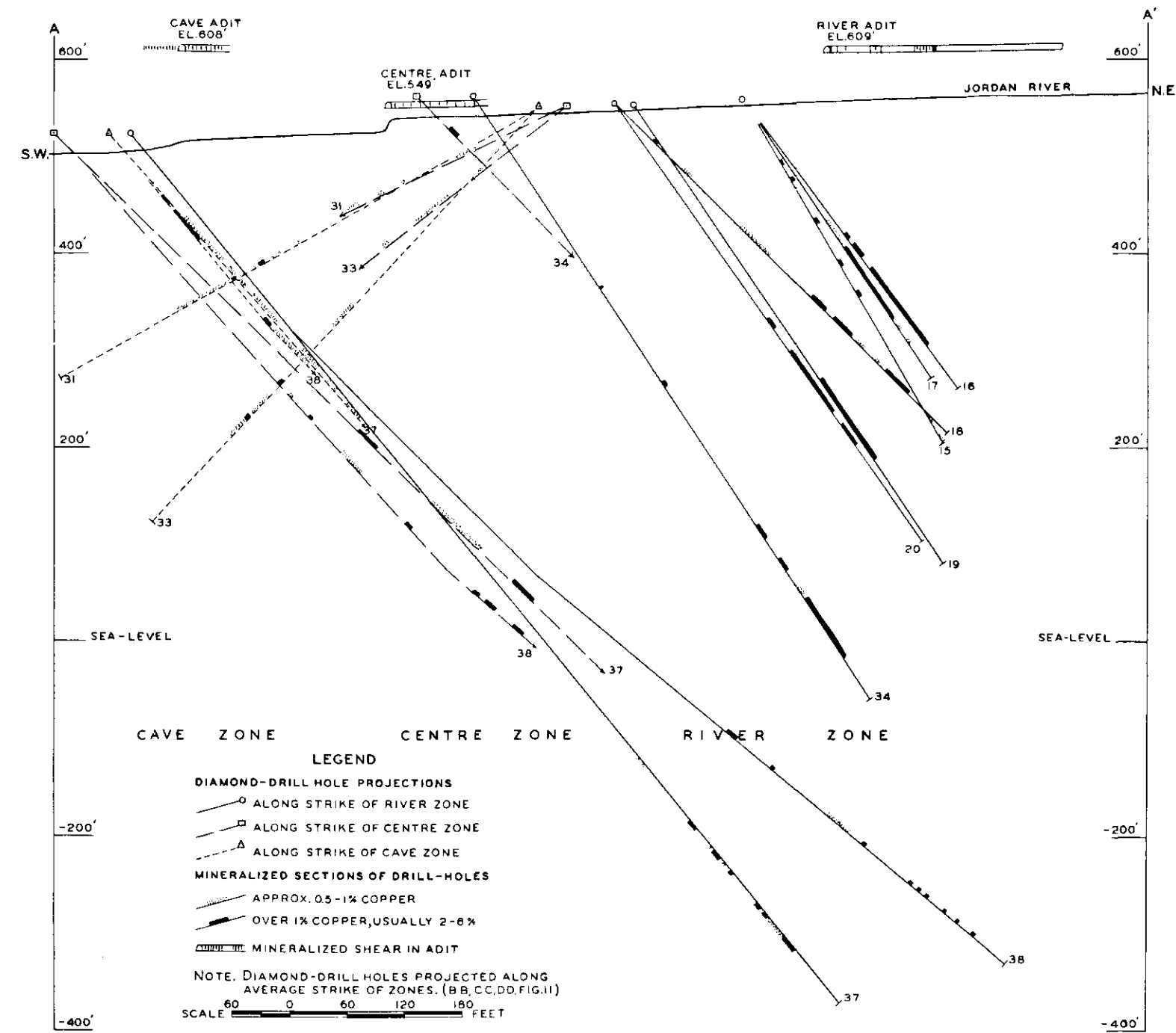


FIG.12. SUNLOCH-GABBRO. VERTICAL SECTION ALONG A LINE BEARING N 60° E (A-A' FIG.11)

Diabase Dykes.—Many dark-green fine- to medium-grained diabase dykes are well exposed, cutting the gabbro in the canyon. The dykes generally trend northwesterly with the strike of the gabbro, but they vary, and some are almost at right angles to this general northwest trend. They range in width from a fraction of an inch to about 10 feet, but widths about 3 feet are the most common. Finely crystalline to glassy selvages, up to a quarter of an inch thick, may be seen along the walls of most of these dykes. Such selvages are characteristic of the walls of quickly chilled dykes. Under the microscope the rock away from the selvages is seen to possess a typically diabasic texture and to consist principally of plagioclase laths and stout hornblende crystals. No dykes have been definitely recognized in the basalt, but, because of their megascopic similarity to the basalt, they might easily be missed, and as some thin sections of diabasic basalt seen in the drill core are very similar in texture and composition to thin sections of known dykes, it is probable that diabase dykes also occur in the basalt.

Faults.—With the exception of the shear zones themselves, which are in all likelihood faults, there are no outstanding faults near the workings. The sharply incised canyon of the river suggests that it may be along a fault, but no evidence of such a fault has yet been found. Numerous slips and shears up to several inches wide may be seen in the adits, where some appear to offset the ore a few feet. None of these faults contain sulphides.

Ore Deposits

The ore deposits are copper-bearing shear zones in basalt, and are principally along the northeast and southwest contacts of the centre band of gabbro. The basalt for several hundreds of feet from the shear zones has been largely replaced by hornblende, and in the shear zones this hornblendized basalt has been mineralized with chalcopyrite, pyrrhotite, and pyrite. The amount of copper in the zones is variable, and not all the zones carry sufficient copper to be of economic interest. The combined gold and silver in the ore have a value amounting to about 50 cents per ton at prices prevailing in 1950, and small amounts of nickel have been reported in pyrrhotite from some of the zones.

Rock Alteration in and Adjacent to the Ore Zones.—The normal gabbro and basalt in the Jordan River-Sooke area contain only minor amounts of hornblende, and the ferromagnesian mineral is relatively unaltered augite. However, in the gabbro and in the basalt, in and near the mineralized zones, the augite has been largely altered to hornblende, and although one may find occasional relicts of augite in the gabbro, it is rare to find any in the basalt. In the same gabbro the plagioclase is largely unaltered, but in the basalt near the workings, much of the plagioclase has been replaced by hornblende.

Under the microscope the rock from the mineralized zones and near by is seen to consist principally of dark-green to light brownish-green hornblende, the augite and plagioclase having been completely replaced. The degree of alteration is spatially related to the ore zones and reaches its culmination in the completely hornblendized rock of the ore zones. The alteration of plagioclase to hornblende appears to be directly proportional to the intensity of hydrothermal alteration and varies not only across the line of strike of the shear zone but, because of lenticularity in width of shearing, also varies along the strike.

Mineralogy.—The sulphides in the ore zones include chalcopyrite, pyrrhotite, pyrite, and small amounts of molybdenite. Chalcopyrite and pyrrhotite are much more abundant than pyrite. Microscopic laths of cubanite have been noted in some specimens of chalcopyrite, and minute blebs or wisps of pentlandite have been seen in pyrrhotite. Much of the pyrite has a striking colloform texture. This texture is present in specimens taken from the adits and also from deeper parts of the drill-holes; therefore, it is not a result of deposition by surface waters. Consequently, such pyrite may be expected in much of the ore, both near the surface and at depth. A small amount of native

copper, as disseminated grains and as a leaf-like coating along short slips, has been seen in core from holes beyond the mineralized zones. Magnetite, in scattered grains, is common not only in the ore zones but also beyond them, and appears to have formed as a rock mineral rather than as a hydrothermal mineral.

The ore-sulphides form a pattern of gash-like veinlets and irregular lenticular masses in the hornblende rock of the shear zones. To the veinlets, which are short and sharp-walled, Young has applied the very appropriate term "crackles."*

Some chalcopyrite veinlets also contain quartz, and from their walls, replacement has proceeded outward so that adjacent stringers coalesce and give rise to irregular masses of replacement sulphide. Where the sulphides are abundant, the associated hornblende rock is likely to be coarser-grained than average. In such places the veinlets of chalcopyrite are often accompanied by a selvage of coarse hornblende crystals about a quarter of an inch long that contrast with the much finer grain of the enclosing hornblende rock. In addition to its occurrence in veinlets and lenses, a small amount of chalcopyrite occurs as disseminated grains. Such chalcopyrite is found throughout a much greater width of shear-zone hornblende-rock than that in which the stringers and lenses of sulphide occur. The amount of copper in the disseminated grains of chalcopyrite is small, and in this particular habit the chalcopyrite is only of mineralogical interest.

The gangue includes, in addition to hornblende, several minerals that occur in small amounts with the sulphides and also form well-defined white stringers, usually only a fraction of an inch wide, both within and beyond the ore zones. Plagioclase is the most abundant mineral in these stringers. The other minerals, listed in decreasing order of abundance, include clinozoisite, chlorite, apatite, calcite, scapolite, and quartz. Small amounts of chalcopyrite are found in some stringers, even beyond the ore zones.

Ore Zones. — Prospecting has disclosed twelve (Fig. 10) zones of chalcopyrite-pyrrhotite mineralization, and although not all contain copper in economic amounts, they will be described under the heading of "Ore Zones." They are within bodies of completely hornblendized rock. The basalt was sheared, perhaps only slightly, but sufficiently to render it susceptible to alteration to hornblende rock and subsequently was further fractured to permit more localized deposition of sulphides. The ore zones grade outward from highly mineralized hornblende to only slightly mineralized rock. The extent to which a shear zone may be ore depends on the number and width of the chalcopyrite-filled fractures and on the amount of massive lenticular chalcopyrite present. Widths of mineralized shear zone, that may be considered ore, range from 3 feet to as much as 100 feet but are usually less than 50 feet. The longer sections of the ore zones, such as along the 400-foot main drift in the River adit, usually range from 1 to 4 feet (Fig. 11).

Four of the twelve ore zones are in basalt close to the northeastern contact of the centre band of gabbro; five are in basalt, close to the southwestern contact of the gabbro; and three are in areas mapped as gabbro. Most of the zones have a general northwesterly trend parallel to the strike of the basalt and the gabbro, but because the ore is cut and displaced by fractures of several different strikes and because the zones have not been fully explored, the trends of the several zones are not all known with certainty.

The ore zones appear to have steep to vertical dips. Deep drilling on the River zone indicates fairly definitely a dip from 70 to 80 degrees southwest, whereas drilling on the Centre and Cave zones suggests more nearly vertical dips for them. Outcrops and drilling on the other zones suggest that they also have steep to vertical dips.

The ore zones from northeast to southwest are those along the northeast contact of the gabbro—River, Centre (Archibald), Cave, and Turnbull; those within the gabbro—Bend, Stewart, and Hornet; those along the southwest contact of the gabbro—Winkler,

* Webster defines "crackles" as "a peculiar cracked surface common in much oriental pottery and porcelain and in some glassware; a condition of the surface of an oil painting, characterized by numerous fine cracks."

Tiger, Yellow Cliff, Robertson, and Caulfield. Three of the zones along the northeast contact—namely, the River, Centre, and Cave—have so far proved to be the most promising. Some drilling has been done on other zones, but most of it has been done and is being continued on the River, Cave, and Centre zones.

River Zone: This zone, at present the most promising of the ore zones, is toward the centre of the Sunloch No. 6 claim. It is in basalt and trends north 30 degrees west, roughly parallel to the northeastern contact of the central band of gabbro and about 1,200 feet from it. As deduced from diamond-drill hole data, this zone appears to dip from 70 to 80 degrees southwest. The zone (Figs. 11 and 12) has been explored by the River adit and by twenty diamond-drill holes—namely, Nos. 2, 3, 7, 8, and 9—drilled prior to 1920, most of them by The Consolidated Mining and Smelting Company, and Nos. 15 to 20, inclusive, Nos. 23 to 25, inclusive, and Nos. 30, 34, 37, and 38 drilled in 1949–50 by Hedley Mascot. Drill-hole No. 39, drilled during the winter of 1950–51, is also directed to intersect the River zone. The zone ranges in width from a foot to about 100 feet and is traceable along its strike for about 1,100 feet. This length includes 520 feet of shear zone followed by the River adit and an additional 570 feet indicated by diamond drilling. An open-cut and diamond drilling indicate that chalcopyrite mineralization extends for at least 200 feet above the River adit. Chalcopyrite mineralization has also been traced to a depth of about 910 feet below the level of the adit by drill-hole intersections that are 150 feet northwesterly from the portal of the adit. This gives a total traced depth of 1,110 feet for chalcopyrite mineralization in the River zone.

Cave Zone: This zone (Figs. 11 and 12) is toward the southwestern corner of the Sunloch No. 6 claim, about 700 feet southwesterly from the River zone. It trends north 40 degrees west and contains widely spaced stringers and lenses of chalcopyrite over a width of about 130 feet. This widely scattered type of mineralization has been traced by the Cave adit for about 500 feet and further extended by drilling for about 120 feet northwesterly from the portal of the adit. Little is known of its vertical extent, but Drill-hole No. 33 intersected chalcopyrite mineralization 200 feet below the level of the adit and Drill-hole No. 37 intersected mineralization under the river 250 feet below the adit. The length now indicated, including the exposures in the Cave adit and intersections in drill-holes under the river, is about 600 feet. However, a northwesterly trending line of old open-cuts and trenches that extends for 500 feet northwesterly from the southwestern corner of the Vulcan No. 6 claim exposes chalcopyrite mineralization that is possibly on the projected northwest extension of the Cave zone and would therefore extend the zone another 900 feet to give a traceable length of 1,500 feet. These workings were dug prior to 1920 and are now badly overgrown and sloughed, and the mineralization only difficultly recognizable. However, some cuts do expose rather abundant chalcopyrite mineralization. Hedley Mascot drilled Holes No. 21 and Nos. 26 to 28, inclusive, to intersect the downward extension of mineralization in the cuts but did not intersect encouraging copper mineralization.

Centre (Archibald) Zone: This zone (Figs. 11 and 12) is on the Sunloch No. 6 claim, 300 feet southeasterly from the River zone. It strikes north 70 degrees west, diagonally to the adjacent River and Cave zones; like the Cave zone, it appears to be vertical. The Centre zone may include a band of mineralization, referred to as the Gordon, that is 50 feet northeasterly from the main locus of mineralization. The Centre zone has been crosscut by the Centre adit, and has been followed by a drift, 100 feet long, in the River adit. It has also been intersected by diamond-drill holes Nos. 31, 33, 37, and 38. As seen in the Centre adit, the zone consists of a 120-foot width of widely spaced stringers of chalcopyrite. As seen in the crosscut and drift in the River adit, the zone has been mineralized by abundant chalcopyrite over a maximum width of 4 feet for a length of about 50 feet. The traced length of chalcopyrite mineralization in this zone, as measured from the Centre adit to the crosscut and drift in the River adit, is

660 feet, and the depth as measured from the Centre adit to the intersection in Drill-hole No. 38 is 320 feet.

Turnbull Zone: This zone is on the Vulcan Nos. 5 and 6 claims. It is reported to have been exposed for about 700 feet in a northwesterly trending line of open-cuts and trenches, about 150 feet southwesterly from the line of old trenches at the northwest end of the Cave zone. The cuts on the Turnbull zone were also dug prior to 1920 and are now completely overgrown and sloughed in and quite unrecognizable. A plan dated 1921 shows assay widths and values in these cuts that range from 0.36 per cent copper over 16.5 feet to 0.75 per cent copper over 6.3 feet. Hedley Mascot drilled four holes—Nos. 21, 22, 26, and 27—to intersect the downward extension of this zone, but failed to find encouraging mineralization.

Bend Zone: This zone is exposed on the north bank of the river about 200 feet downstream from the mouth of Robertson Creek. Unlike most of the other zones, it is within an area mapped as gabbro and appears to strike north 65 degrees east. This zone was found in 1949 by Hedley Mascot, who drilled Hole No. X-6 (Fig. 10) to intersect its downward extension at the river's edge. It is reported that the hole had intersected encouraging chalcopyrite mineralization before the ground became too difficult to drill with the light machine that was being used. The mineralization in this zone is traceable for 200 feet on the surface and to a depth of 40 feet.

Stewart Zone: This zone is on the Vulcan No. 3 claim on the southeast side of the river and canyon; it is about 2,000 feet southerly from the Cave zone. The work on this zone consists of several open-cuts and strippings dug about 1922. These workings are less overgrown than those on the northwestern side of the river, and the mineralization is more readily seen. No drilling has been done on this zone. The mineralization consists of moderate amounts of pyrrhotite and lesser amounts of chalcopyrite, over a width of 30 feet, and is traceable northerly in cuts for about 250 feet. The length as at present traceable in the cuts is about 250 feet in a northerly direction. Although these workings are in an area mapped as underlain by gabbro, they are in hornblende rock that is similar to that accompanying the shear zones in basalt beyond the gabbro, and it may be that the Stewart zone, like most of the other zones, is in an area of basalt.

Hornet Zone: This zone is near the northwest corner of the Black Hornet claim and has been prospected by some old strippings and a 12-foot adit dug about 1920 in the banks of Sinn Fein Creek about 1,700 feet upstream from the mouth of the creek. No diamond drilling has been done on the zone. It is close to the southwestern contact of the gabbro band but appears to be well within the general area of gabbro. However, the mineralization is in hornblende rock similar to that derived from basalt, and it may therefore be part of an inclusion or embayment of altered basalt within the gabbro. A moderate amount of chalcopyrite was seen in stringers in one open-cut and in the adit.

Southwest Contact of the Gabbro: The Winkler, Tiger, Yellow Cliff, Robertson, and Caulfield zones are in basalt. They are close to and trend along the southwest contact of the central band of gabbro. These zones were originally prospected by surface workings, but some of them have been recently diamond drilled by Hedley Mascot. The copper mineralization disclosed so far has not proved economic, and work on the showings has been suspended for the present.

Winkler Zone: This zone, about 10 feet wide, is on the Vulcan No. 2 claim, about 2,500 feet southwesterly from the Cave zone. It has been prospected by an adit 83 feet long on the east or upper side of the old railway grade and by two strippings about 50 feet below the grade; this work was done about 1920. In 1929 British Metals Corporation drilled a hole below the railway grade that intersected chalcopyrite mineralization in a short section about 100 feet below the level of the adit. In 1949 Hole No. X-5 (Fig. 10), drilled from the southeast bank of the river, intersected chalcopyrite mineralization at 40

feet. Although the position of this mineralization, 175 feet below and 320 feet northwesterly from the adit, is not quite along the strike as projected from the adit and stripings, nevertheless it may be along the downward extension of the Winkler zone.

Tiger Zone: This zone is on the Tiger claim, about 400 feet southwesterly from the Winkler zone. It has been prospected by stripings that extend intermittently from the railway grade northwesterly down to the river. Chalcopyrite mineralization belonging to this zone has been intersected in Holes Nos. 35 and 36 (Fig. 10) drilled by Hedley Mascot. The mineralized zone ranges in width from a few feet to 20 feet. Its traceable length is about 300 feet and its traced vertical extent, as measured from the railway grade to the intersections in Holes Nos. 35 and 36, is about 360 feet.

Yellow Cliff Zone: This zone, trend northwesterly, crosses the boundary between the Black Hornet and Tiger claims near the south bank of the river. It has been traced for 120 feet by pits dug by Hedley Mascot in 1949. The pits expose mineralization over a width of 8 feet. Although no drilling has been done on this zone, it appears to be one of the most promising of those along the southwestern contact of the gabbro.

Robertson Zone: This zone, not seen by the writer, is reported to trend northerly close to the line between the Black Hornet and Hornet Fraction claims. The only working reported to be on it is an open-cut, now completely overgrown, on the north bank of the river. The zone is reported to be about 12 feet wide in this cut.

Caulfield Zone: This zone is in the southeastern corner of the Black Hornet claim, directly across the river from the Yellow Cliff zone. It was prospected about 1920 by a small amount of stripping that is now badly overgrown. In 1949 Hedley Mascot drilled four short X-ray holes, Nos. X-1 to X-4, inclusive (Fig. 10), but the difficulty of finding suitable set-ups for the drill militated against obtaining conclusive results. It has been suggested that exposures of copper mineralization in the bed of Sinn Fein Creek, about half-way along the west boundary of the Black Hornet claim, may be the northwesterly continuation of the Caulfield zone. The Sinn Fein exposures are about 1,200 feet from those on the river.

Ore Tonnages

Development up to the present time, particularly the 1949-50 diamond drilling done by Hedley Mascot, has indicated ore of commercial widths and grades in the River zone below the river. By that drilling the company has outlined a composite block, consisting of three smaller blocks, measuring from 150 to 300 feet in length, from 75 to 100 feet in width, and extending to a depth of about 850 feet below the river. They estimate that these blocks may contain about 600,000 tons of ore of milling grade, containing copper and a little gold.

Deep drilling done by the company across the Centre and Cave zones suggests that below the river these zones may also contain considerable additional tonnage, but of somewhat lower grade than the ore in the River zone.

Biogeochemical Studies

Because of the widespread interest in the application of biogeochemistry in prospecting for base metals, the writer made limited biogeochemical studies on the Sunloch-Gabbro properties in the summer of 1950. The general technique of biogeochemical prospecting is based on the quantitative determination of minor or trace elements in plant or tree growth. The method used is a colorimetric, dithizone neutral mixed-colour-end-point method, as described by Warren and Delavault (1949, p. 538) and by White (1950, pp. 368, 369). For the work at Jordan River, equipment and procedures used were similar to those described by White (1950, p. 369).

The writer and one assistant spent about one week in August, 1950, collecting tree samples on the Sunloch and Gabbro properties. The work was entirely exploratory, and

no attempt was made to make a comprehensive coverage of the properties. However, the results obtained did demonstrate the value of the method, in that the trees sampled indicated the presence or absence of copper mineralization in the underlying bedrock.

The tree cover on the ground near the Sunloch and Gabbro workings is extremely variable. On the canyon slopes above the railway grade, the trees are principally large hemlocks, 1 to 2½ feet in diameter at the base and several tens of feet tall. These trees lack branches that can be reached from the ground. Smaller trees are practically absent. It was therefore impossible to collect samples in such areas. However, because suitable trees are abundant along the two old railway grades, one connecting the River adit with the Cave adit and the other extending about a mile downstream from the Cave adit, this belt was selected for sampling. Although this belt is narrow, it has the advantage of crossing the local structure, and it includes the principal mineralized shear zones and the intervening unmineralized rock.

The soil along this "traverse line" is thin and consists principally of black muck a few inches deep. Glacial drift is found only as a few scattered pockets in protected depressions.

As the soil cover is thin, many of the tree roots extend completely through it into crevices in the underlying rocks. Therefore, one would expect the source of the metals in the tree growth, drawn from the soil and surface water in the soil or crevices in the underlying bedrock, to be extremely local. This appears to be so because high results were obtained from trees found only a few feet from trees giving low results.

As trees measuring from 2 to 4 inches at the base are abundant along the railway grades, this size-range was selected for sampling. For purposes of comparison and also as a check on the sampling, it was decided wherever possible to collect samples from a balsam and from a hemlock adjacent to it. Such a sample-pair included two trees no more than 5 feet apart; the results obtained indicate that this was a safe distance. The pair was given a number, the balsam being designated by "A" and the hemlock by "B" (e.g., 3A and 3B). The same number of twigs was taken from each tree and from about the same relative positions on each tree. The branches from which the twigs were broken were spaced completely around the tree and from 4 to 6 feet above the ground. Only the growth of the previous two years, as shown by nodes on the twigs, was taken.

Each sample consisted of about thirty twigs stripped of their needles and weighed about 5 grams before ashing. The analyses were made the same day the twigs were collected, and the results calculated to parts per million (p.p.m.) of fresh undried samples. The reagents used and the procedure followed are as detailed by White (1950, p. 369).

Forty-nine sample-pairs were taken, and the amounts of copper and zinc, in parts per million (p.p.m.) of fresh twigs, were determined, and copper:zinc ratios calculated for each sample. This information is summarized in the table on page 185, and the approximate position of each sample-pair is indicated in Figure 10. In the table the samples are grouped according to the mineralized zone, or unmineralized rock, over which they were taken.

A study of the table indicates that the metal values and the copper:zinc ratios are higher in groups of samples taken over the mineralized shear zones than in groups of samples taken over the intervening unmineralized rock.

The tree samples from the River zone contained less copper and zinc than those from the other zones. The writer ascribes this condition to the many bare rock bluffs and the thinness of the soil in which the only trees available for sampling on the zone were found to be growing. Samples taken from the Centre, Cave, Winkler, and Tiger zones showed higher than average metal values and copper:zinc ratios. Samples from the intervening greenstone and gabbro showed on the whole lower values and ratios than those in the mineralized zones.

Within any one group, individual samples may differ widely from the average for the group. Sample No. 18, in a group of samples taken from a section of unmineralized greenstone, is higher than the average for its group; at this sample point, it is possible that the trees may have been growing above a minor shear carrying disseminated chalcocopyrite.

The high values in Sample No. 28 are difficult to explain. The trees sampled were larger than average, 8 inches at their base, and grew at the edge of a gravel pit, about 15 feet deep, half a mile from any known ore zone. Rock outcrops are entirely absent in this area, and underlying geology is unknown. It is possible that these higher than average metal values reflect an underlying orebody, but it is also possible that the higher values may be due to either the larger size of the trees as compared to those sampled along the old railway grade, or to the considerable depth of glacial material in which they were growing. Because of these differences in tree size and overburden, the values in Sample No. 28 should not be compared with those in the other samples in the table. The sampling of trees of similar size and growing in similar overburden should be extended in several directions from the gravel pit to determine whether the values of Sample No. 28 are anomalous or quite general for the area.

An interesting feature of the results is the close correspondence between the variation in metal content of adjacent balsam and hemlock.

Although it was not possible to grid-sample a large area on the Sunloch and Gabbro properties, sampling along a traverse, which crosses the principal mineralized shear zones and intervening on mineralized rock, has shown that the metal content of the trees reflects the presence or absence of important mineralization in the underlying bedrock. For the biogeochemical method to be of value in prospecting for extensions of ore zones on these properties, a type of plant growth that is uniformly distributed over the whole area to be prospected should be selected. The writer would suggest that the following tree or plant growth might be used: The bark of large hemlock-trees, a core obtained by boring into the trunks of large hemlock, or the leaves of the shrub salal (*Gaultheria shallon*). Unfortunately, wood and dead bark of trees have not proved too suitable elsewhere (Warren and Delavault, 1949, p. 541). Although the variation in metal content of salal leaves is at present unknown, the writer would suggest that, because it is so common in the coastal regions of the Province, some exploratory geochemical studies should be made using this shrub.

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